Demographics and Covid-19: Linear Regression vs. K-means Clustering Approaches

Lily Campbell

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1 Abstract

In this paper I examine the effects of demographic features on Covid 19 case and death rate by utilizing K-means clustering, Principle Component Analysis, and Linear Regression. I found that higher proportions of people of color in a county correlates with a higher rate of Covid-19 related cases and deaths. In addition, I found that Linear Regression is better able to handle data with many outliers.

2 Introduction

I have studied methods for analysis of the effect of demographics on Covid-19 cases and deaths because I believe it may be helpful to learn how the magnitude of the inequities across socioeconomic status and race can be revealed through analysis of this health crisis. John's Hopkins University has released their data on the daily total of Covid-19 cases and deaths by county for the US. Because counties serve as a good proxy for socioeconomic groups, and the US census provides detailed information on the demographics of each county, I am able to link county death rate to county demographics. "Demographic clustering" is used frequently in economics and marketing to understand consumer groups. Demographic data gives discrete data points, and as such may be clustered. Linear regression is also used frequently to determine the demographic makeup of populations which contract infectious disease (see references), and as such I will be comparing K-means clustering and Linear Regression.

3 Background

K-means Clustering

K-means is a common clustering algorithm, which uses distance from group center points to categorize data points into the desired number of clusters. After random initialization of centroids, the euclidean distance from each data point

to the centroids is measured, and each data point is assigned to the cluster it is closest to. The mean of all the data points assigned to each cluster is taken, and the centroid is updated to the mean point. This is done iteratively until the change between previous and updated centroids is small.

PCA
Principle Component Analysis (PCA) is used to determine the directions which maximize variance of data. By projecting onto the principle components which represent a high amount of the variance of the data, analysis may be done on a lower dimension than the original data.

Linear Regression
Linear Regression is a method to find a line (or hyper-plane) of best fit for a set of data. By finding the least squares of residuals, the line or plane of best fit is found. The equation of the plane of best fit reveals the coefficient on each feature, and the statistical significance of each feature's impact may be analyzed.

4 Method

To find which demographic features impact Covid-19 case and death rate most, I compare the results of three algorithms, K-means Clustering, K-Means Clustering with PCA, and Linear Regression.

K-means Clustering
I use K-means clustering to sort the counties into groups by demographic features. I then find the mean value of each feature for the clusters of counties, and the mean Covid-19 case and death rate for each cluster. I can then compare the rate of cases and deaths to the mean of the demographic features for each cluster in order to identify correlation between demographic features and Covid-19 rates.

K-means Clustering with PCA
In order to reduce noise captured in the analysis of the 17 demographic features, I used PCA to reduce the dimension of the data and then projected onto the first two principle components. The first two principle components represent 99% of the variance of the data, thus I found it sufficient to only project onto the first two components. I then did K-means clustering of the projected data, and analyzed those clusters.

Linear Regression
In order to confirm my findings from analysis of the K-means clusters, I used Linear Regression. I regressed cases per capita and deaths per capita on each of the demographic features to see which had statistically significant impacts.
Cases per Capita $= \beta_0 + \beta_1 IncomePerCapita + \beta_2 \%Black + \beta_3 \%Hispanic +$

$\beta_4 \%White + \beta_5 \%Asian + \beta_6 MeanCommute + \beta_7 UnemploymentRate + residual$

5 Findings

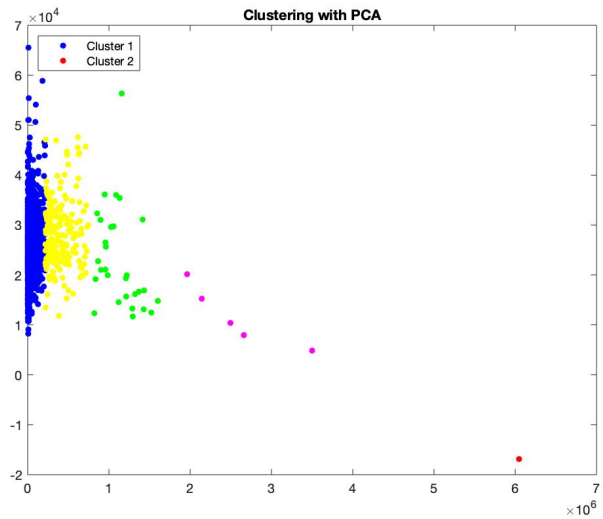
K-means Clustering
I ran trials clustering the data into 3, 5 and 7 clusters, but found that increasing the amount of clusters did not affect the outcome. Thus I will report my findings with 5 clusters.

	Cluster 1	Cluster 2	Cluster 3	Cluster 4	Cluster 5
Mean Cases per Capita	0.9303	1.5490	2.7884	1.8533	1.8554
Mean Deaths per Capita	1.2647e-04	2.3403e-04	8.3851e-04	2.8959e-04	4.1066e-04
Mean %Black Citizens	8.5091	10.6600	16.4724	8	13.5655
Mean %Hispanic Citizens	8.2163	32.7000	24.5759	48.2000	15.4730
Mean Income per Capita	23838	30795	31602	28337	31264

Mean cases and deaths per capita rank evenly throughout each cluster (e.g. mean cases and mean deaths are lowest in cluster 1, highest in cluster 3, etc.), which is to be expected. While there is some inconsistency, clusters with high rates of Covid-19 related cases and deaths correspond with clusters which have higher portions of people of color in their population. This confirms my intuition that due to institutional racism, health care provided in counties with high proportions of people of color is less able to treat and prevent Covid-19 cases than other counties.

Income per capita has a positive correlation with cases and deaths per capita. This result is somewhat surprising, as I did not expect that counties with a high income per capita would correspond with counties which have a high rate of deaths from Covid-19. However this may be because counties which have high income per capita are also high in population density, and thus may be more prone to cases of Covid-19.

K-means Clustering with PCA
When the data is projected onto its first two principle components, the clusters may easily be seen in two dimensions.



The demographic features of clusters when projected onto the first two principle components have a slightly stronger correlation with the rate of cases and deaths:

	Cluster 1	Cluster 2	Cluster 3	Cluster 4	Cluster 5
Mean Cases per Capita	0.9290	1.8533	1.8257	2.7884	1.5490
Mean Deaths per Capita	1.2609e-04	2.8959e-04	4.0128e-04	8.3851e-04	2.3403e-04
Mean %Black Citizens	8.5022	8	13.4011	16.4724	10.6600
Mean %Hispanic Citizens	8.1795	48.2000	15.6630	24.5759	32.7000
Mean Income per Capita	23821	28337	31133	31602	30795

Results from first performing PCA then clustering the projected data are approximately the same as the above results from performing K-means clustering directly on the demographic data, however the results are somewhat more consistent as noise is removed when working with the principle components. The graph of the data projected onto the first two principle components reveals a weakness of K-means clustering, in that it is ill-equipped to handle data with many outliers. Cluster size varied greatly as outliers were put into very small clusters, while the majority of counties were only divided into two clusters. I tried to combat this by making demographic features per capita to reduce outliers created by population size, but the make up of a minority of cities (which also make up most of the outliers) have greatly different demographics than rural or suburban counties. I did not want to remove these outlier counties however, as they represent large cities and some Covid-19 hot spots.

Linear regression
I used STATA to perform regressions on my data.

Below are the most significant results of running the regression: Cases per Capita $= \beta_0 + \beta_1 IncomePerCapita + \beta_2 \%Black + \beta_3 \%Hispanic + \beta_4 \%White + \beta_5 \%Asian + \beta_6 MeanCommute + \beta_7 UnemploymentRate + residual$

```
. *regression cases per capita on demographic features
.
. reg Casespercapita IncomePerCap Hispanic Black White Asian MeanCommute Unemplo
> ymentrate

      Source |       SS           df       MS      Number of obs   =     3,139
-------------+----------------------------------   F(7, 3131)      =     51.62
       Model |  .018055375         7  .002579339   Prob > F        =    0.0000
    Residual |  .156449802     3,131  .000049968   R-squared       =    0.1035
-------------+----------------------------------   Adj R-squared   =    0.1015
       Total |  .174505178     3,138   .00005561   Root MSE        =    .00707

------------------------------------------------------------------------------
Casespercap~a |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
 IncomePerCap |   8.55e-08   2.83e-08     3.03   0.002     3.01e-08    1.41e-07
     Hispanic |   .0000515   .0000188     2.75   0.006     .0000147    .0000883
        Black |    .000127    .000018     7.07   0.000     .0000918    .0001622
        White |  -.0000256   .0000175    -1.46   0.144    -.0000599    8.72e-06
        Asian |   .0001598   .0000601     2.66   0.008      .000042    .0002775
  MeanCommute |   .0000443   .0000253     1.75   0.081    -5.38e-06    .0000939
Unemploymen~e |  -2.14e-06   .0000491    -0.04   0.965    -.0000985    .0000942
        _cons |   .0010354   .0018811     0.55   0.582    -.0026529    .0047236
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Results from regressing cases per capita on the demographic features show that there are statistically significant positive impacts from proportion of Hispanic, Black and Asian citizens on cases per capita. Mean Distance of Commute has a positive impact on the amount of cases per capita. I found this result to be surprising as a higher mean commute seems to indicate lower population density, however gentrification and urban spread tend to increase the commute and population in areas with lower quality health infrastructure surrounding cities.

Below are the most significant results from the regression: Deaths per Capita $= \beta_0 + \beta_1 IncomePerCapita + \beta_2 \%Black + \beta_3 \%Hispanic + \beta_4 \%White + \beta_5 \%Asian + \beta_6 MeanCommute + \beta_7 UnemploymentRate + residual$

```
. *regression of death per capita on demographic features
.
. reg Deathspercapita IncomePerCap Hispanic Black White Asian MeanCommute Unempl
> oymentrate

      Source |       SS           df       MS      Number of obs   =     3,139
-------------+----------------------------------   F(7, 3131)      =     73.78
       Model |  .000063562         7  9.0803e-06   Prob > F        =    0.0000
    Residual |  .000385349     3,131  1.2308e-07   R-squared       =    0.1416
-------------+----------------------------------   Adj R-squared   =    0.1397
       Total |  .000448911     3,138  1.4306e-07   Root MSE        =    .00035

------------------------------------------------------------------------------
Deathsperca~a |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
 IncomePerCap |   1.52e-08   1.40e-09    10.83   0.000     1.24e-08    1.79e-08
     Hispanic |   7.36e-07   9.31e-07     0.79   0.430    -1.09e-06    2.56e-06
        Black |   7.06e-06   8.91e-07     7.92   0.000     5.31e-06    8.81e-06
        White |  -9.75e-07   8.69e-07    -1.12   0.262    -2.68e-06    7.29e-07
        Asian |   4.15e-06   2.98e-06     1.39   0.164    -1.69e-06    1.00e-05
  MeanCommute |   5.05e-06   1.26e-06     4.02   0.000     2.59e-06    7.52e-06
Unemploymen~e |   7.08e-06   2.44e-06     2.90   0.004     2.30e-06    .0000119
        _cons |  -.0003917   .0000934    -4.20   0.000    -.0005747   -.0002086
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When examining Covid-19 related deaths, unemployment rate becomes a statistically significant result, while it was not significant when examining cases per capita. This may suggest that a county may expect a portion of Covid-19 cases to be terminal if its unemployment rate is high, perhaps due to lower quality of preexisting social services.

6 Conclusion

When analyzing the effects of demographics on Covid-19 related cases, Linear Regression was more successful. By examining the line of best fit given by Linear Regression, I am able to see the individual and combined effects of each demographic feature on Covid-19 death and case rate. When analyzing demographic groups identified by K-means clustering, the magnitude of each individual demographic feature's effect was difficult to determine, I could only identify the results of the combined effect of the demographic features.

In addition, Linear Regression was better able to handle data with many outliers. While outliers may disproportionately bias a line (or hyper-plane) of best fit, Linear Regression is able to produce results which include outlier data points in the analysis.

7 References

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